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Optimal Strategy for Energy-Efficient Management of Greenhouse Climate Control

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ABSTRACT In this research, we present a multi-objective optimization model for efficient energy management of greenhouse (GH) microclimate control systems. The study aims to minimize energy costs and peak demand while controlling the GH temperature, humidity, and CO_2 levels within a predefined range. A mathematical model is developed to quantify the power consumption of GH components, including lighting, heating, cooling, CO_2 injection, and ventilation systems, considering electricity costs based on time-of-use (ToU) tariffs distinguishing peak and off-peak periods. Multi-objective convex optimization, implemented in MATLAB, optimizes control decisions for these systems over 24 hours in 15-minute intervals. The decision variables, ranging from 0 (off) to 1 (fully on), determine the operational states of each system. The model uses historical weather data from Regina, Canada, on July 1st (as a sample day), applied to a 1000 m^2 GH in the simulation. To verify the feasibility and test the effectiveness of the proposed control and optimization method, also to provide a sensitivity analysis based on changes in weather elements, including temperature, solar radiation, and wind speed, we perform the proposed strategy over the first day of each month for every 12 months of the year. Tomato cultivation is chosen for GH due to its compatibility with the controlled environment (CO_2, relative humidity (RH), and temperature). The findings reveal a remarkable up to 43.13% energy savings a day with the optimized strategy compared to the base scenario of the GH control system. The results illustrate optimal activation patterns for GH systems throughout the day, offering a practical tool for smart GH energy management.

INDEX TERMS Energy management, climate control, cost-effectiveness, greenhouse heating system, multi-objective optimization.

I. INTRODUCTION

A. BACKGROUND AND RESEARCH PROBLEM STATEMENT

ENERGY management, optimization, and control are important topics that have attracted much attention in both the industrial and agricultural sectors [1], [2]. With the growth of the world population and the increase in the demand for agricultural products, the importance of productivity in agricultural production and efficient energy management in greenhouse (GH) systems has become more evident. GHs, as one of the effective solutions to increase agricultural product production under different weather conditions, require accurate and efficient energy management [3]. GH climate control systems, including lighting, heating, cooling, ventilation, and injection of carbon dioxide CO_2, play an important role in maintaining optimal conditions for plant growth. However, the high energy consumption and operational costs of these systems are significant challenges for GH operators [4]. GH operations are under increasing pressure to optimize energy use amid rising costs and environmental concerns, particularly for climate control systems that regulate temperature, lighting, and CO_2 levels. Therefore, in this paper, a multi-objective optimization model is presented for efficient energy management of GH climate control systems, which reduces energy costs and peak demand during a typical day, while maintaining an indoor temperature. By integrating real weather data and employing convex optimization (CVX) techniques, the study develops a practical framework to determine optimal operating states for all involved energy components. This approach not only reduces operating expenses compared to conventional GH control strategies but also provides a scalable tool for GH operators to make informed and energy-efficient decisions, contributing to sustainable agricultural practices. To accurately present the topic of optimization and control in smart GHs, we should consider several important items, including the following factors:

1- Different Structures of GHs: GH structures are generally divided into two main categories: traditional GH and modern GH. Traditional GHs mainly use materials such as glass and plastic for covering and have simpler climate control systems. These types of GHs usually have natural ventilation systems that rely on natural airflow to regulate temperature and humidity. In contrast, modern GHs use more advanced technologies, such as automatic temperature, humidity, and ventilation control systems that provide energy efficiency and more precise control. These systems usually include sensors and intelligent controllers that continuously monitor environmental data and make optimization decisions [5].

2- Energy Management in GHs: Energy management in GHs includes the optimal use of energy resources to reduce costs and increase productivity. This management can include the use of efficient heating and cooling systems, optimal natural and artificial light, and the optimization of ventilation. The energy efficiency in GH can also be achieved through the use of renewable energy sources (RES), such as solar energy and geothermal heating systems [6]. Studies have shown that the use of new technologies such as intelligent control systems can help reduce energy consumption and operating costs. For example, the use of solar panels to provide part of the energy required by GHs can lead to a reduction in dependence on fossil energy sources and a reduction in costs [7].

3- Demand Response (DR) and Peak Demand: DR and peak demand management are of particular importance in GH, as these two factors play a vital role in optimizing energy consumption and reducing operating costs. Peak demand refers to hours or periods when energy consumption is at its highest, and energy costs are usually higher in these periods. Peak demand management through DR, which involves regulating and controlling energy consumption based on actual demand and weather conditions, can help reduce pressure on the power grid, as well as lower energy costs [8]. This process involves the use of intelligent and automatic systems to optimally adjust the heating, cooling, lighting, and ventilation devices to always maintain a balance between supply and demand. In this way, GHs can create more sustainability and productivity in their production by taking advantage of DR and peak demand management, in addition to reducing costs [9].

4- Smart Control Methods: Intelligent control methods in GH include the use of algorithms and automatic systems to adjust various parameters such as temperature, humidity, and ventilation. These methods can include the use of machine learning systems, optimization algorithms, and smart sensors [10]. By simulating evolutionary processes in nature, optimization algorithms seek optimal combinations of control parameters that lead to a reduction in cost and a greater increase in productivity.

B. LITERATURE REVIEW

Some methods in the literature have provided control and optimization strategies in GHs. For example, the paper [7] presented a novel hierarchical control method and mathematical optimization models for GHs that can be easily integrated into energy hub management systems (EHMS) in smart grids to optimize the performance of their energy systems. The goal was to minimize total energy costs while maintaining critical GH parameters such as indoor temperature and humidity, CO_2 concentration, and light levels within acceptable limits. The work [11] presented a novel central control method for a smart GH network with a microgrid (MG) that optimized energy management considering renewable energy resources (RES) and weather conditions. The proposed method allowed central control to manage the temperature and production of GH products using an optimization model with limited prediction and real-time sensor information. In addition, the authors of the article [12] presented an energy management method for GHs with RES that minimized energy consumption, considering the randomness of these sources and the weather conditions outside. By modeling the thermal and electrical processes of the GH in two fast and slow time frames, this method reduced computational complexity and provided an optimal solution for energy consumption. In the paper [13], the authors proposed a strategy for optimal GH energy management in MG, considering both the connection to the grid and the island mode (based on internal resources). In island mode, where resources are limited, priority is given to various control parameters to promote better plant growth. In addition, an algorithm is provided to prevent these parameters from going out of the acceptable range in consecutive time intervals. Another study [14] proposed a GH automation system based on the Internet of Things (IoT), which saved energy and resources by better controlling temperature, humidity, light and irrigation and ultimately increased the final product. By optimally controlling functions based on real-time data, this system reduced operating costs and increased the profitability of the GH. The authors of the paper [15] presented a method for optimal energy management of smart GHs using model predictive control (MPC) in a cluster MG to reduce energy consumption while maintaining high quality. This system worked in harmony using a central controller, RES, energy storage, and power exchange with the main grid. In another study [16], researchers examined a solution to optimize energy consumption in GHs equipped with solar panels in rural areas of China. Using this method, a GH with an area of 3,996 square meters with 25% coverage of solar panels can save up to 15% in energy costs. In addition, the work [17] proposed a sustainable smart GH model utilizing the Artificial Bee Colony (ABC) algorithm to minimize energy consumption while ensuring optimal plant comfort. By optimizing key environmental factors (temperature, humidity, CO_2, and sunlight), the ABC-based system, governed by a fuzzy controller, demonstrated better energy efficiency and plant comfort compared to other optimization algorithms like GA, FA, and ACO, highlighting its potential for reducing operational costs in modern agriculture. Furthermore, in [18], by building a mathematical model of GH energy consumption, the study presented a method to predict energy consumption and also optimized the daily temperature of GH to reduce energy consumption. The final goal of the study was to achieve production with high efficiency, high product quality, and low energy consumption. Another study [19] proposed an improved data-driven MPC framework, utilizing an Artificial Neural Network (ANN), to manage energy and temperature in high-tech GHs, especially in arid climates. The framework used a double-layer approach to effectively account for system uncertainties and maintain precise microclimate control. Results showed this method not only achieved better temperature accuracy, but also significantly reduced energy utilization by up to 20.01% compared to existing control systems.

The work in [20] addressed the challenges of fluctuating energy prices and transmission congestion by proposing a DR model for energy users. The authors have used a cooperative game pricing method to allocate congestion costs and applied the alternating direction method of multipliers to decouple the integrated energy system, ensuring privacy and independence. The results in a power grid system illustrated that the proposed algorithm converged and effectively reduced user participation costs in DR. In addition, another research [21] presented a dispatch strategy for an integrated agricultural energy system, taking into account a ladder-type carbon trading mechanism and DR to optimize energy use. The model balanced energy production with carbon emissions trading to minimize costs while maintaining system reliability. The results demonstrated that the presented approach effectively reduced carbon emissions and energy costs, improving the overall efficiency of the system. In addition, the work [22] proposed a stochastic model for coordinated planning of the expansion of generation and transmission, addressing uncertainties of high-penetration RES and incorporating DR programs. The chance-constrained method ensured flexibility in adapting to different levels of DR, helping delay investments and save costs. Case studies revealed that even a small proportion of DR can significantly reduce investment costs by more than 20%, validating the effectiveness of the model. Table 1 shows a comparison of recent related works on GH optimization, control, and energy management in terms of applications, strengths, and weaknesses.

C. MOTIVATION AND MAIN CONTRIBUTIONS

This paper tackles the challenges head-on by introducing a smart and integrated approach to energy optimization in GH systems. Our main goal is to connect energy efficiency with environmental control, two objectives that have often been treated separately in past research. By using advanced multi-objective optimization techniques, we aim to reduce energy costs and peak demand while still maintaining the ideal internal climate conditions for optimal plant growth. Unlike many earlier studies that focus on individual components of GH systems or view optimization as a one-dimensional problem, our research takes a broader, systems-level perspective. We optimize several interrelated factors such as temperature, humidity, lighting intensity, ventilation rates, and CO_2 levels, all of which are crucial for plant health and energy use. This comprehensive approach allows for dynamic trade-offs between competing goals, creating a truly flexible energy management framework. The application of the CVX tool used in our work is renowned for its ability to tackle convex and linear optimization challenges; it offers the adaptability needed to address real-world scenarios that are intricate and constrained. By integrating time-of-use (ToU) energy tariffs and peak load restrictions into our model, we ensure our framework aligns with actual utility pricing, enhancing its practical application. Additionally, we aimed to provide a detailed analysis of energy consumption patterns over time, backed by high-fidelity MATLAB simulations. This allows us to assess the effectiveness of different parameter settings and show how strategic changes can lead to significant improvements in energy efficiency and cost savings. In summary, the contributions of this paper can be summarized as follows:

- We design a comprehensive mathematical representation and develop a convex optimization model for efficient energy management of GH microclimate control systems, to minimize both total energy cost and the peak electricity demand of key GH subsystems—lighting, heating, cooling, ventilation, and CO_2 injection—under real-world climatic data.
- We develop a scalable, real-time capable convex program by integrating dynamic GH physics, ToU tariffs, and peak constraints into a DCP-compliant model, solved efficiently using CVX.
- We implement the proposed optimization model using 15-minute time steps over a 24-hour horizon and extend the analysis to 12 representative days (one day per each month) to evaluate annual performance and feasibility.
- We validate the model through a GH case study, based on cold region real climate data, focusing on tomato cultivation with realistic environmental constraints on temperature 20–30°C (ideal growing range for diverse plant crops), CO_2 and relative humidity (RH).
- We demonstrate that the optimized control strategy achieves an average daily energy saving of 27.55% compared to the base control scenario, up to 43% or more cost and energy saving, highlighting superiority, substantial economic and operational benefits.
- We provide a flexible and scalable optimization framework adaptable to various GH sizes, crop types, and climate conditions, contributing to the advancement of smart and sustainable agricultural energy management.

TABLE 1. A comparison of recent quality works on GH optimization, control, and energy management.

Ref.	Application	Main Contribution	Strengths	Weaknesses
[17]	Smart GHs with AI control for ventilation/lighting	Artificial Bee Colony algorithm for energy minimization and plant comfort index maximization	Achieves 15–25% energy reduction in simulations; handles multi-parameter optimization (temp, humidity, CO_2)	Computationally intensive for real-time use; no peak demand or ToU tariff integration
[23]	Zero-energy GHs with IoT sensors	Systematic review of smart tech for near-zero consumption, emphasizing data-driven control	Maps 2018–2023 trends; proposes IoT frameworks reducing emissions by 40%	Broad scope dilutes depth; ignores convex optimization for scalability and peak shaving
[24]	Thermal management in heating/cooling systems	Review of passive/active technologies for energy-efficient GH envelopes	Comprehensive coverage of materials (phase-change materials); identifies 20–30% savings potential	Lacks optimization framework; focuses on design rather than real-time operation or multi-objective trade-offs
[25]	Smart GH control via IoT and DEVS simulation	DEVS-based discrete-event modeling of climate, irrigation, and energy devices for optimal scheduling	Reduces energy by 37.12% in simulation; avoids mathematical optimization; evaluates multiple configurations	No real-time optimization or convexity; simulation-only; no ToU tariffs or peak demand constraints
[26]	Crop-specific GH cultivation	Review of control algorithms (PID, MPC) for low-energy operation	Analyzes yield-cost trade-offs; recommends hybrid controls for 10–20% efficiency gains	Descriptive, no new model; overlooks CO_2 and RH in energy models, no convexity guarantee
[27]	IoT-enabled GHs with WSNs	WSN-based monitoring for IEQ and energy optimization in green buildings	Integrates sensors for real-time IEQ feedback; 25% energy savings in pilot tests	Focuses on residential analogs; scalability issues for large-scale GHs without renewables or peak constraints
[28]	Renewable integration in agricultural facilities	AI for forecasting and optimization in solar/wind GH systems	Reviews ML models reducing CO_2 by 30–50%; forward-looking on edge AI	General to RES; lacks GH-specific dynamics like ventilation/CO_2 injection or multi-objective peak management

Our work offers not only a theoretical contribution but also a practical solution to a pressing agricultural challenge. By intelligently aligning energy use with plant needs and electricity price conditions, this research helps create greener, more resilient food production systems.

The rest of this paper is organized as follows: the proposed control and optimization strategy is presented in Section II; the obtained results of the control system and optimization, as well as a comparison between strategies, are illustrated in Section III. In Section IV, we discuss and analyze the results and, finally, we conclude the paper in Section V.

II. METHODOLOGY

The GH energy management model optimizes temperature, CO_2 concentration, and RH to maintain ideal plant growth conditions while minimizing energy cost and peak demand. This is achieved through a dynamic energy balance and mass balance framework.

A. ENERGY BALANCE

The GH's internal temperature, T_{GH} (K), is governed by the energy balance in Equation (1), which balances the thermal mass of the air with the net energy flows.

$$\rho.V.c_P\frac{dT_{GH}(t)}{dt} = Q_{in}(t) - Q_{out}(t) \tag{1}$$

where ρ represents the air density (kg/m^3), c_P gives the specific heat capacity of the air $(J/kg\acute{}K)$, T_{GH} represents the temperature inside the GH (K), and $\frac{dT_{GH}}{dt}$ denotes the rate of change of temperature inside the GH over time (K/s). In addition, $Q_{in}(t)$ and $Q_{out}(t)$ are the rates of energy (W) entering and leaving the GH at time t, respectively; ρ represents the air density (kg/m^3), and V denotes the GH volume (m^3). We must quantify the energy inputs and outputs affecting the internal environment to maintain the GH temperature within the desired range. The total energy input rate, Q_in, encompasses heat gains from various sources, including solar radiation, heating systems, lighting, CO_2 injection, ventilation, and occupants. These contributions are calculated using Equation (2), with each term detailed in subsequent equations.

$$Q_{in}(t) = Q_{sun}(t) + Q_{heat}(t) + Q_{light}(t) + Q_{CO_2}(t) + Q_{vent}(t) + Q_{pp}(t) \tag{2}$$

where $Q_{sun}(t)$, $Q_{heat}(t)$, $Q_{light}(t)$, $Q_{CO_2}(t)$, $Q_{vent}(t)$, and $Q_{pp}(t)$ are the energy input or gain (W) from solar radiation, heating, lighting, CO_2 injection, and inlet air of ventilation systems at time t, respectively. In addition, the energy output rate of the system must also be determined. Formula (3) calculates the output energy or loss rate (Q_{out}) for all elements involved in the GH building.

$$Q_{out}(t) = Q_{cool}(t) + Q_{wall}(t) + Q_{win}(t) + Q_{nv}(t) + Q_{soil}(t) + Q_{deh}(t) + Q_{eva}(t) \tag{3}$$

where $Q_{cool}(t)$, $Q_{wall}(t)$, $Q_{win}(t)$, $Q_{nv}(t)$, $Q_{soil}(t)$, $Q_{deh}(t)$, and $Q_{eva}(t)$ are the energy loss (W) through the chilling pipe, wall, window, natural ventilation, soil and dehumidification in the GH at time t respectively. In this regard, we need to calculate all these gains and losses. For calculating the heat gain from sunlight (Q_{sun}), Equation (4) is given.

$$Q_{\text{sun}}(t) = A_{\text{glass}} \times I_{\text{solar}}(t) \times \tau_{\text{glass}} \tag{4}$$

where A_{glass}, I_{solar} and τ_{glass} are the area of the glass, the intensity of solar radiation, and the transmittance of the glass, respectively. To obtain the heat gain from heating system $(Q_{heat}(t))$, Equation (5) is given.

$$Q_{heat}(t) = \dot{m}_f(t) \times c_{pf} \times (T_{\text{inlet,h}}(t) - T_{\text{outlet,h}}(t)) \tag{5}$$

where $\dot{m}$ represents the mass flow rate of the fluid in the pipes and c_{pf} denotes the specific heat capacity of the fluid. In addition, $T_{outlet,h}$ and $T_{inlet,h}$ are the temperature (K) of the input and output of the heating system. Furthermore, we can calculate the heat gains from the lighting system, CO_2, and people using Equations (6) to (8) [29].

$$Q_{\text{light}}(t) = P_{\text{light}}(t) \times \eta_{\text{light}} \tag{6}$$

$$Q_{\text{CO}_2}(t) = \dot{m}_{\text{CO}_2}(t)\, c_{p,\text{CO}_2}\, \left(T_{\text{CO}_2,\text{inj}}(t) - T_{\text{air}}(t)\right) \tag{7}$$

$$Q_{\text{pp}}(t) = N_{\text{pp}} \times Q_{\text{pers}}(t) \tag{8}$$

where P_{light} and η_{light} are the power input to the lighting system and the efficiency of the lighting system converting electrical energy to heat, respectively. Also, $\dot{m}_{CO_2,max}$ (kg/s) is the maximum CO_2 injection rate, c_{p,CO_2} (J/kg·K) represents the CO_2 specific heat, and δT_{CO_2} (K) denotess the temperature difference. In addition, N_{pp} and Q_{pers} represent the number of people and the heat gain per person (depending on activity level), respectively. The heat gain from the ventilation system is calculated using Equation (9).

$$Q_{vent}(t) = \rho \times c_{pa} \times \dot{V}_{vent} \times (T_{out}(t) - T_{in}(t)) \tag{9}$$

where ρ represents the density of air, T_{out} and T_{in} are the ambient and GH temperatures, respectively. $\dot{V}_{vent}$ also denotes the volumetric flow rate of ventilated air.

In addition, to obtain heat loss from the cooling system (Q_{cool}), Equation (10) can be used.

$$Q_{cool}(t) = \dot{m}_f \times c_{pf} \times (T_{\text{outlet,c}}(t) - T_{\text{inlet,c}}(t)) \tag{10}$$

where, $T_{outlet,c}$ and $T_{inlet,c}$ are the temperatures (K) of the input and output of the cooling system. The losses of walls and windows can be obtained using the Equations (11) and (12) too.

$$Q_{wall}(t) = U_{wall} A_{wall} (T_{in}(t) - T_{out}(t)) \tag{11}$$

$$Q_{win}(t) = U_{win} A_{win} (T_{in}(t) - T_{out}(t)) \tag{12}$$

where U_{wall} and U_{wain} are the overall heat transfer coefficients ($W/m^2.K$) of the walls and windows, respectively. Also, A_{win} and A_{win} are the surface area (m^2) of the walls and windows, respectively. To obtain heat losses from natural ventilation (Q_{nv}), Equation (13) is given [30].

$$Q_{\text{nv}}(t) = \rho \times c_{pa} \times \dot{V}_{\text{nat}} \times (T_{\text{in}}(t) - T_{\text{out}}(t)) \tag{13}$$

where $\dot{V}_{\text{nat}}$ is the volumetric flow rate of natural air. We can calculate the heat losses through soil (Q_{soil}) using Equation (14) [31].

$$Q_{\text{soil}}(t) = k_{\text{soil}} \times A_{\text{soil}} \times \frac{(T_{\text{surf}}(t) - T_{\text{depth}}(t))}{L_{\text{soil}}} \tag{14}$$

where k_{soil} and A_{soil} are the thermal conductivity of the soil and the cross-sectional area through which heat is transferred, respectively. T_{surf} and T_{depth} are the surface and depth temperatures of the soil. L_{soil} is also the depth over which the temperature difference occurs.

The heat removed by dehumidification (Q_{deh}) and evaporation (Q_{eva}) can be obtained using Equations (15) and (16) [30].

$$Q_{\text{deh}}(t) = \dot{m}_{\text{air}} \times c_{pa} \times (T_{\text{in}}(t) - T_{\text{deh}}(t)) + \dot{m}_{\text{wat}} \times h_{\text{lv}} \tag{15}$$

$$Q_{\text{eva}}(t) = \dot{m}_{\text{wat}}(t) \times h_{\text{lv}} \tag{16}$$

where $\dot{m}_{\text{air}}$ denotes the mass flow rate of air; T_{in} and T_{deh} represent the inside temperature and temperature after dehumidification, respectively. Also, $\dot{m}_{\text{wat}}$ and h_{lv} are the mass flow rate of condensed water and the latent heat of vaporization, respectively [32]. However, to control and provide an exact amount of humidity rate in the GH, we will provide the strategy to calculate of RH dynamics and rates in the rest of the study. In the study, we are required to calculate values over a specific time interval, so the simulation is implemented over 15-minute intervals ($\Delta t = 0.25$ hours) for a 24-hour period, using sensor-based and optimized control strategies. We can obtain a time step using Eq. (17).

$$\Delta t = \frac{T}{N} \tag{17}$$

where in this case, $N = 96$ and $T = 24$ are the number of intervals and the total time (hours) used in the calculation and simulation, respectively. To calculate the indoor GH temperature (T_{GH}) over time t, we can model the thermal dynamics of the GH temperature based on the principle of energy balance using Equation (18).

$$\begin{aligned} T_{GH}(t) = T_{GH}(t-1) + \frac{\Delta t}{\rho V c_p}[&Q_{sun}(t) + Q_{heat}(t) \\ &+ Q_{light}(t) + Q_{CO_2}(t) + Q_{vent}(t) + Q_{pp}(t) \\ &- Q_{cool}(t) - Q_{wall}(t) - Q_{win}(t) - Q_{nv}(t) \\ &- Q_{soil}(t) - Q_{deh}(t) - Q_{eva}(t)] \end{aligned} \tag{18}$$

Equation (17) models the $T_{GH}(t)$ considering initial temperature $(t-1)$ and the net energy change over a time step Δt. V is the volume of GH (m^3). The term $\frac{\Delta t}{\rho V c_p}$ converts the net power (W) into a temperature change (K) taking into account the thermal mass of the air ($\rho V c_p$). Each Q term represents an energy flow, with positive terms increasing the temperature and negative terms decreasing it. It should be noted that T_{GH} should be maintained subject to the specified range between the minimum (T_{GH}^{min}) and maximum (T_{GH}^{max}) limits as given in Equation (19).

$$T_{GH}^{min}(t) \le T_{GH}(t) \le T_{GH}^{max}(t), \quad \forall t = 1, \ldots, 96 \tag{19}$$

The next stage is to calculate the airflow rate (m_{air}) in (kg/s) of the GH building. Equation (20) calculates the airflow inlet for GH ventilation.

$$m_{air} = \rho A_v v \tag{20}$$

where m_{air}, A_v, and v are the mass flow rate of air (kg/s), the cross-sectional area of ventilation openings (m^2), and the velocity of air (m/s), respectively.

B. CO_2 AND HUMIDITY DYNAMICS

The GH climate, beyond temperature, is shaped by the dynamic behavior of CO_2 concentration and humidity, both modelled via mass balance principles over 15-minute intervals. The energy usage time interval is also 15 minutes (Δt=0.25 hours) over 96 intervals (24 hours). CO_2 concentration (ppm) is modeled using a simplified mass balance given in Eq. (21).

$$CO_2(t) = CO_2(t-1) + \Delta t \left(k_{CO_2} s_{CO_2}(t) - k_{vt} s_{vent}(t)\right) \tag{21}$$

subject to the following constraints:

$$CO_2^{min}(t) \leq CO_2(t) \leq CO_2^{max}(t), \quad \forall t = 1, \ldots, 96 \tag{22}$$

$$0 \leq s_{CO_2}(t) \leq 1, \quad \forall t = 1, \ldots, 96 \tag{23}$$

where $k_{CO_2} = 50$ ppm/h and $k_{vt} = 10$ ppm/h are empirical rates for CO_2 injection and ventilation loss. Also, $CO_2^{min} = 500$ ppm and $CO_2^{max} = 700$ ppm. Additionally, to calculate the RH (%), we can use Eq. (24).

$$RH(t) = RH(t-1) + \Delta t \left(k_{vt,RH} s_{vt}(t) - k_{cl,RH} s_{cool}(t)\right) \tag{24}$$

subject to:

$$RH_{min}(t) \leq RH(t) \leq RH_{max}(t), \quad \forall t = 1, \ldots, 96 \tag{25}$$

$$0 \leq s_{cool}(t) \leq 1, \quad \forall t = 1, \ldots, 96 \tag{26}$$

where $k_{vt,RH} = 0.1$ (%/h), $k_{cl,RH} = 0.05$ (%/h), are empirical coefficients in the RH dynamics model, representing the rate of RH increase per hour due to ventilation and decrease due to cooling, respectively, in the GH simulation. Also, the optimal range for the RH ratio for tomato growth should be $RH_{min} = 40$ (%) and $RH_{max} = 60$ (%).

C. OPTIMIZATION PROBLEM

Energy consumption for each component (heating, cooling, lighting, CO_2 injection, ventilation) over Δt is:

$$E_n(t) = P_n s_n(t) \Delta t \tag{27}$$

where $n \in \{heat, cool, light, CO_2, vent\}$. P_n is the nominal power ratings (W) of each n, and $s_n(t)$ are the decision variables, subject to the following constraints:

$$0 \leq s_n(t) \leq 1, \quad \forall t = 1, \ldots, 96 \tag{28}$$

Therefore, the total energy consumption ($E_{tot}(t)$) is calculated using Eq. (29).

$$\begin{aligned} E_{tot}(t) &= \sum_{n=1}^{5} P_n s_n(t) \cdot \Delta t = E_{heat}(t) + E_{cool}(t) \\ &\quad + E_{light}(t) + E_{CO_2}(t) + E_{vent}(t) \end{aligned} \tag{29}$$

The energy cost accounts for ToU tariffs are computed by Eq. (30).

$$OF_1 = C_{energy} = \sum_{t=1}^{N} cost(t) \cdot \frac{E_{tot}(t)}{1000} \tag{30}$$

where $cost(t)$ (\$/kWh) is constrained by:

$$cost(t) = \begin{cases} 0.18 & 64 \leq t \leq 80 \text{ (16:00–20:00)} \\ 0.14 & \text{otherwise} \end{cases} \tag{31}$$

The peak demand, which refers to the highest energy consumption point from all components over t, is obtained using Eq. (32).

$$OF_2 = D_{peak} = \max_{t \in [1,\ldots,N]} \left[\sum_{n=1}^{5} P_n s_n(t) \right] \tag{32}$$

An important point is to normalize the energy cost and peak demand, making both objectives dimensionless and comparable. So we have both new objectives in the form of:

$$\widehat{OF}_1 = \frac{OF_1}{C_{\text{ref}}} \tag{33}$$

$$\widehat{OF}_2 = \frac{D_{peak}}{D_{\text{ref}}} \tag{34}$$

where C_{ref} and D_{ref} are reference values used to normalize the energy cost and peak demand, respectively. In this study, C_{ref} and D_{ref} represent the baseline (unoptimized) daily energy cost and peak demand of the GH under standard operation. This normalization ensures that the weighted-sum objective function (f) is unitless, allowing a fair trade-off between minimizing both objectives. To improve optimization of energy consumption and peak demand in the GH system, weights (w_1) and (w_2) are introduced into the multi-objective function to balance trade-offs between minimizing the total energy cost (C_{energy}) and peak demand (D_{peak}). The weighted objective function (f) is:

$$f = w_1 \widehat{OF}_1 + w_2 \widehat{OF}_2 \tag{35}$$

The weights w_1 and w_2 are varied to obtain the Pareto front that demonstrates the compromise between cost and demand minimization. The best w_1 and w_2 are chosen in such a way that they can minimize the final daily operational cost. This weighted approach allows for prioritization based on operational or economic goals, combining the objectives into a single scalar function. Therefore, we can minimize the final objective function (f) using Equation (36).

$$\begin{aligned} &\min_{s_n(t)} f \\ &\text{subject to :} \begin{cases} 0 \leq s_n(t) \leq 1 \\ T_{GH}^{min}(t) \leq T_{GH}(t) \leq T_{GH}^{max}(t) \\ CO_2^{min}(t) \leq CO_2(t) \leq CO_2^{max}(t) \\ RH_{min}(t) \leq RH(t) \leq RH_{max}(t) \leq 1 \\ cost(t) = \begin{cases} 0.18 & 64 \leq t \leq 80 \\ 0.14 & \text{otherwise} \end{cases} \end{cases} \\ &\quad \forall t = 1, \ldots, 96 \\ &\quad n \in \{heat, cool, light, CO_2, vent\} \end{aligned} \tag{36}$$

The problem is formulated using disciplined convex programming (DCP) rules and solved via CVX, a MATLAB-based convex optimization package. This set of equations has been compiled to optimize energy consumption in GHs and includes calculating the total energy consumption for all devices, determining the cost of energy, calculating the peak demand points, and determining a multi-criteria objective function. The purpose of these equations is to provide a comprehensive framework for optimal energy management in GHs so that, by reducing energy costs and improving the efficiency of the GH system, a suitable environment can be provided for plant growth.

D. JUSTIFICATION OF CONVEX LINEAR APPROXIMATIONS

GH climate dynamics are inherently nonlinear, involving coupled differential equations for temperature, humidity, CO_2, and crop responses, often leading to non-convex optimization challenges [5]. For instance, radiative heat transfer, vapor pressure deficits, and photosynthetic rates introduce quadratic or transcendental terms, as highlighted in Gaussian process-based stochastic MPC approaches for GH control [33]. While such models capture fine-grained interactions, they incur high computational costs, making them less suitable for day-ahead scheduling in practical operations. Our model adopts linear approximations of energy and mass balances (Eqs. (1)–(18)) to ensure convexity and scalability. Key simplifications include:

- Constant coefficients: Heat transfer (U-values), efficiencies (η_{light}), and empirical rates (k_{CO_2}, k_{vt} are held fixed over 15-min intervals (Δt h), valid within the controlled ranges (T, RH, and CO_2).
- Linearized gains/losses: Solar input (Q_{sun}), conduction (Q_{wall}, Q_{win}), and ventilation (Q_{vent}) use first-order terms, neglecting higher-order effects like temperature-dependent emissivity.
- Discrete-time integration: Eq. (18) approximates continuous dynamics for short horizons.

These approximations yield a convex linear program (LP), solvable in <1 sec using CVX/SDPT3 for a 96-interval horizon—orders of magnitude faster than non-convex methods.

III. IMPLEMENTATION AND SIMULATION RESULTS

We have two different scenarios to implement for GH control, including the base scenario and the optimal scenario. The base scenario, known as the sensor-based scenario, implements a rule-based control strategy that independently manages the GH environmental parameters (temperature, CO_2, RH, and lighting) using predefined thresholds, serving as a baseline for comparison with the optimized scenario. The control algorithm employs a logic with binary (on/off) control actions for each component. Each parameter is controlled separately using independent if-else logic, without coordinating trade-offs, in contrast to the multi-objective optimization approach. In our proposed smart strategy, in addition to controlling all the values inside the GH, the smart energy management addresses the ToU of use of electricity price and intelligently reduces energy consumption of the GH's different subsystems and components. Also, we present the results obtained from the simulations performed for multi-criteria energy optimization in the GH. We compare the optimal scenario simulation results with those of the sensor-based scenario GH control system, confirming that, by using a multiobjective optimization algorithm, we can achieve an improvement in the energy and cost performance of GH systems. In the following, the results obtained from the simulation are presented, which clearly show the optimized performance of the systems and their effects on energy consumption and related costs.

TABLE 2. Parameters and values for GH energy management model.

Variable	Unit	Value	Variable	Unit	Value
ρ	kg/m^3	1.2	U_{wall}	$W/m^2 \cdot K$	0.5
V	m^3	4000	A_{wall}	m^2	800
c_{pa}	$J/kg \cdot K$	1005	A_{win}	m^2	1200
T_{GH}	K	293–303	$\dot{V}_{nat}$	m^3/s	0.5
A_{glass}	m^2	800	k_{soil}	$W/m \cdot K$	1.0
I_{solar}	W/m^2	Weather	A_{soil}	m^2	1000
τ_{glass}	–	0.8	T_{surf}	K	295
$\dot{m}_f$	kg/s	0.5	T_{depth}	K	290
c_{pf}	$J/kg \cdot K$	4186	L_{soil}	m	1.0
T_{inlet}	K	313(h), 283(c)	$\dot{m}_{air}$	kg/s	1.2
T_{outlet}	K	303(h), 293(c)	T_{deh}	K	293
η_{light}	–	0.9	$\dot{m}_{wat}$	kg/s	0.001
$\dot{m}_{CO_2}$	kg/s	0.005	A_v	m^2	10
c_{p,CO_2}	$J/kg \cdot K$	850	ν	m/s	Weather
ΔT_{CO_2}	K	5	P_{heat}	W	80,000
N_{pp}	–	2	P_{cool}	W	80,000
Q_{pers}	W	100	P_{light}	W	8,000
V_{vent}	m^3/s	15.0	P_{CO_2}	W	500
T_{out}	K	Weather	P_{vent}	W	2000
T_{in}	K	293–303	-	-	-

A. PLANT SELECTION AND ENVIRONMENTAL PARAMETERS

The GH control system maintains CO_2 concentrations between 500 and 700 ppm, RH between 40% and 60%, and a minimum 25% intensity of lighting during nighttime hours (22:00–5:00). These conditions are optimized for tomato cultivation, which thrives at CO_2 levels of 500–700 ppm to improve photosynthesis, with 650 ppm targeted during the day. The RH range of 40–60% prevents fungal diseases while supporting tomato transpiration. The 25% minimum provides a baseline photosynthetic photon flux density (PPFD) critical for maintaining metabolic activity overnight, reducing energy consumption compared to full lighting while meeting plant physiological needs. The suggested environmental parameters and values during the day and night for tomato cultivation are obtained from studies [34], [35], [36], [37], [38]. However, these controlled items, such as temperature, RH, and CO_2 levels, can be used for many other GH crops. Crops such as strawberries, bell peppers, and cucumbers also need a similar environment to tomatoes in the GH to grow.

B. IMPLEMENTATION PARAMETERS

The parameters and values for the GH energy management model are given in Table 2. The parameters in Table 2 are chosen to support a 24-hour simulation of a 4000 m^3 GH in Regina, Canada, based on July 1st, integrating hourly weather data for temperature, solar radiation, and wind speed. One of the main objectives of this research is to control the GH temperature to be stable in an acceptable range between 20°C and 30°C, which is suitable for most crops. We should

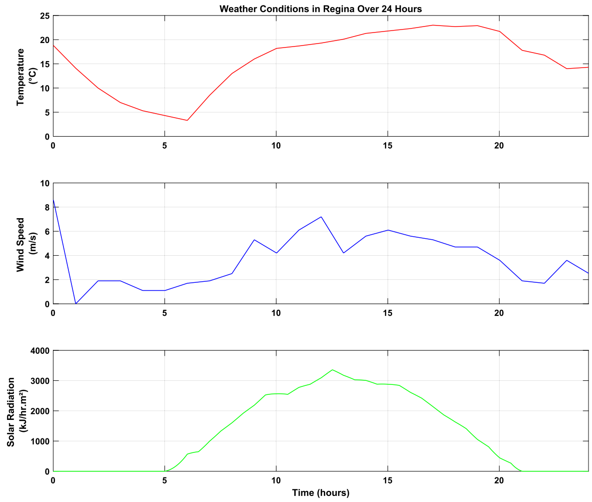

FIGURE 1. Historical and real weather conditions data in Regina for 24 hours on July 1st.

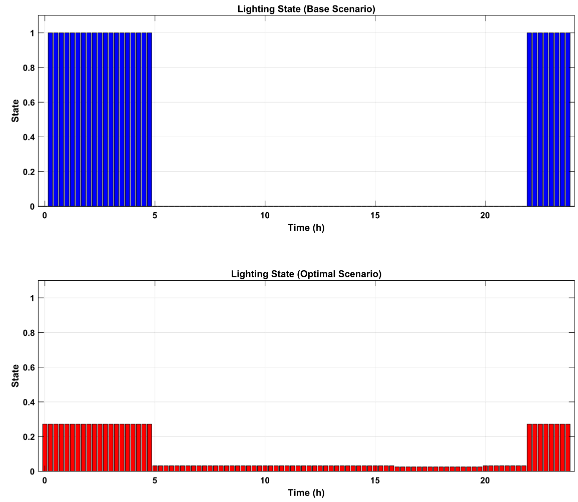

FIGURE 2. The base scenario and optimal scenario states of the lighting system's operation over 24 hours on July 1st.

mention that the control system includes a 2°C deadband with set minimum and maximum limits. Weather conditions have a great effect on the change in GH temperature throughout the day. Fig. 1 shows the real and historical weather conditions data (temperature, wind speed, and solar radiation) in the city of Regina for 24 hours on July 1st, which we used in the simulation. Considering the area and volume of the GH, as well as improving the control system performance and response time of the heating system, we use a 15-minute interval for simulation. Using 15-minute intervals in the simulation can provide better performance compared to, for instance, 1-hour intervals because it allows for finer resolution and more precise control over the GH environment. This higher resolution enables more accurate tracking and response to changes in temperature, ventilation, and other factors, leading to better optimization of energy consumption and cost. In addition, it helps to capture short-term fluctuations that might be missed with longer intervals, resulting in a more responsive and efficient control strategy.

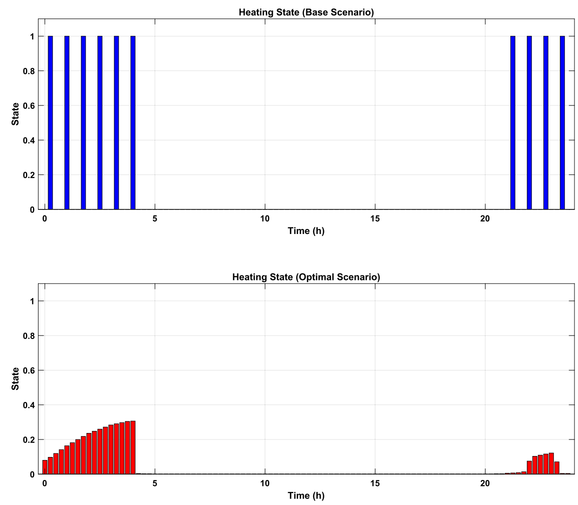

FIGURE 3. The base scenario and optimal scenario states of the heating system's operation over 24 hours on July 1st.

C. OPTIMIZED LIGHT SCHEDULE

Fig. 2 shows the scheduled light states over 24 hours for both the base and optimal scenarios. The optimal value of the light mode is represented by a value of 15 minutes, which can work between '1' (fully active at maximum capacity) and '0' which indicates that it is off. The schedule demonstrates that the lighting system is predominantly activated during the early morning and late evening hours, which is in line with typical plant photoperiod requirements. By limiting light usage during peak daylight hours, optimization ensures that energy is conserved when natural sunlight is sufficient, thereby reducing operational costs.

D. OPTIMIZED HEATING SCHEDULE

Fig. 3 compares the base and optimized scenario states of the heating system's operation over 24 hours, where the heating is most active during cooler times and early morning hours. This strategy helps maintain a stable indoor GH building temperature during periods of lower external temperatures, ensuring that the GH remains within the optimal temperature range for plant growth. The activation of the heating system is minimized during warmer daylight hours, contributing to energy savings by leveraging natural heat sources.

E. OPTIMIZED COOLING SCHEDULE

Fig. 4 compares the optimized and base scenario states of the cooling system's operation. The cooling system is active primarily during the peak afternoon hours when the outdoor temperatures are highest. This activation pattern is crucial for preventing overheating within the GH, which could adversely affect plant health. By concentrating cooling efforts during the hottest parts of the day, the system effectively mitigates

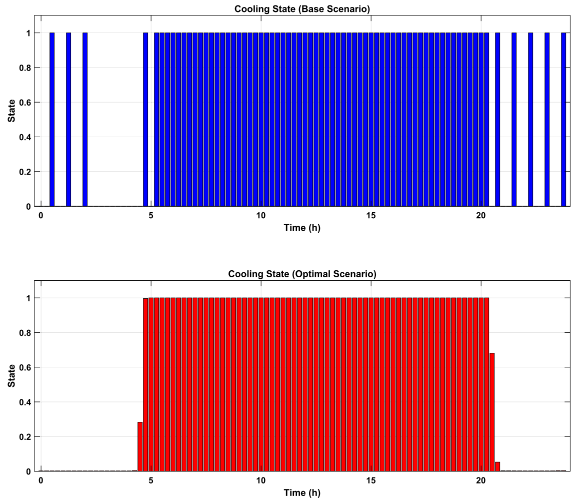

FIGURE 4. The base scenario and optimal scenario states of the cooling system's operation over 24 hours on July 1st.

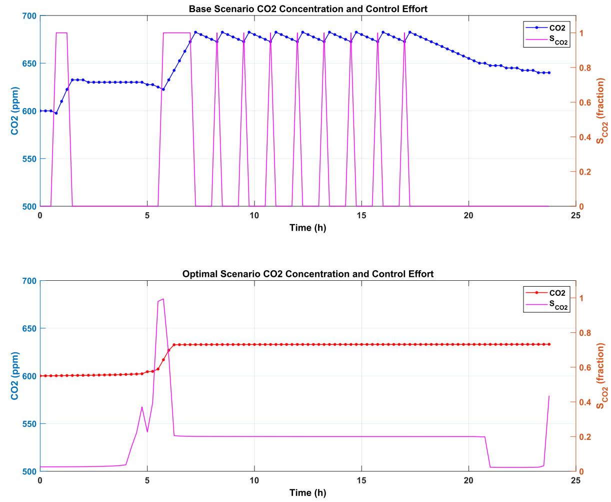

FIGURE 6. The base and optimal CO_2 concentration levels and control signal.

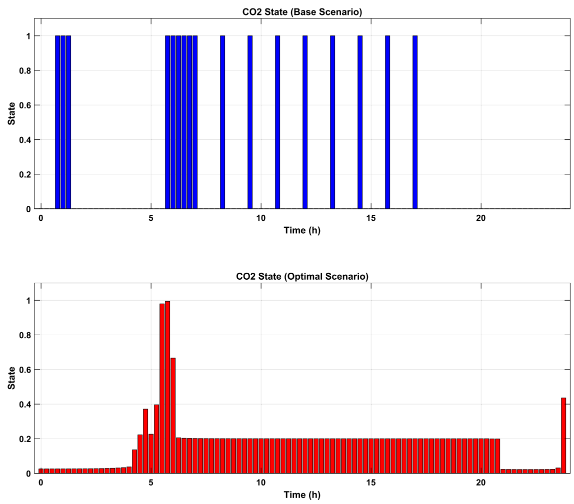

FIGURE 5. The base scenario and optimal scenario states of the CO_2 injection system over 24 hours on July 1st.

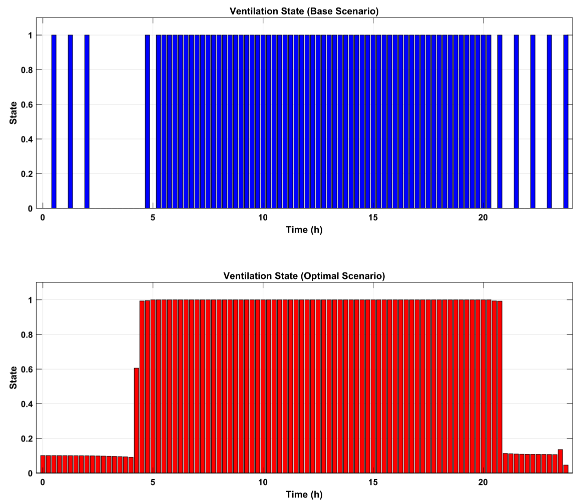

FIGURE 7. The base scenario and optimal scenario states of the ventilation system's operation over 24 hours on July 1st.

thermal stress on plants while optimizing energy use during periods of lower energy costs.

F. OPTIMIZED CO_2 SCHEDULE

Fig. 5 compares the base and optimal scenario states of the CO_2 injection system, which is strategically activated during daylight hours when photosynthesis is most active. Optimization ensures that CO_2 levels are increased when plants can use it best, improving growth while minimizing unnecessary CO_2 usage during nonphotosynthetic periods. This approach contributes to energy efficiency and to the promotion of optimal plant health. To provide an accurate comparison between sensor-based CO_2 concentration and control effort and optimized strategy, Fig. 6 compares the states of both scenarios, as well as the level of the CO_2 (ppm) injection system.

G. OPTIMIZED VENTILATION SCHEDULE AND RH LEVEL

Fig. 7 compares the base and optimal states of the operation of the ventilation system over 24 hours, designed to regulate air exchange in the GH. The schedule indicates increased ventilation during the warmer hours of the afternoon, complementing the cooling system by improving air circulation. This strategy helps maintain appropriate humidity levels and prevents excessive moisture build-up, which is critical for disease prevention and overall plant health. To provide an accurate comparison between sensor-based RH concentration and optimized strategy, Fig. 8 compares the RH levels of both scenarios over 24 hours.

H. TEMPERATURE CHANGES OVER A DAY

Fig. 9 compares the base and optimal scenario temperatures within the GH and the ambient temperature over 24 hours.

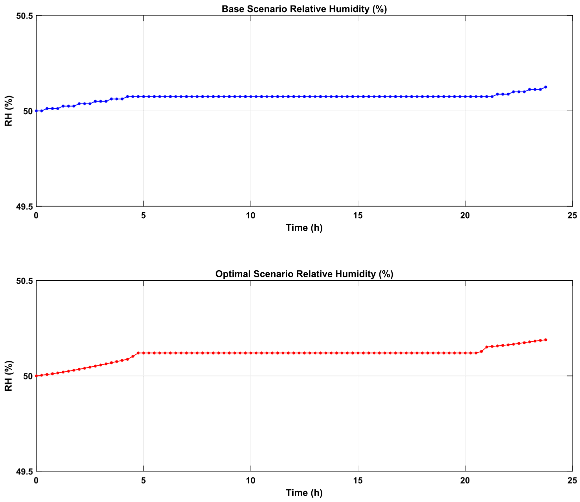

FIGURE 8. The RH levels of the base and optimal scenarios over 24 hours on July 1st.

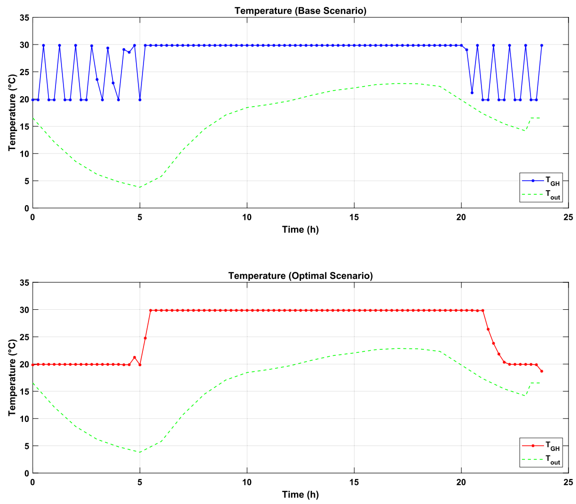

FIGURE 9. The base scenario and optimal scenario temperatures within the GH and the ambient temperature over 24 hours on July 1st.

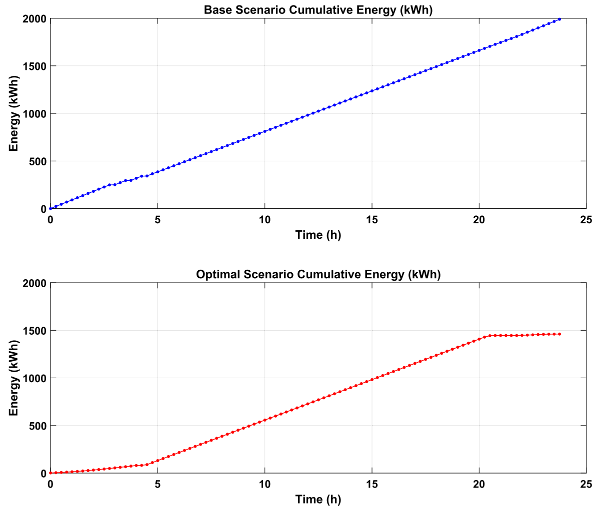

FIGURE 10. The base scenario and optimal scenario cumulative energy consumptions over 24 hours on July 1st.

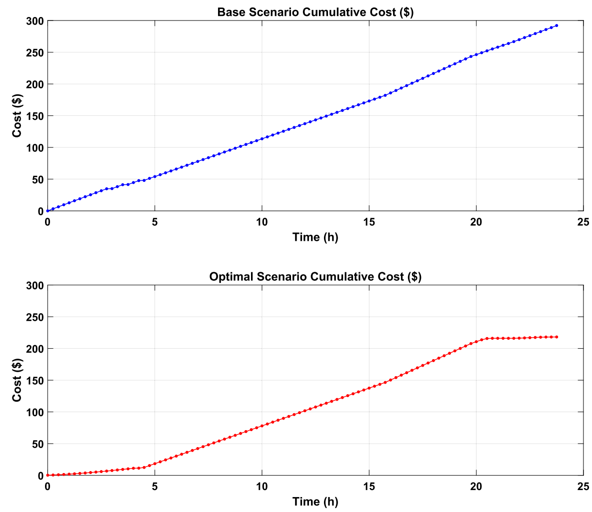

FIGURE 11. The base scenario and optimal scenario cumulative electricity costs over 24 hours on July 1st.

The indoor GH temperature profile closely follows the trends in outdoor temperature, but is moderated by the heating and cooling systems to maintain an optimal range for plant growth. The GH temperature remains relatively stable, fluctuating between 20°C and 30°C, which is the desired range for most GH crops. This stability is achieved through the strategic operation of the heating and cooling systems, as evidenced by their respective schedules.

I. ENERGY AND COST ANALYSIS

Considering both optimal and base scenarios, as well as peak and off-peak demand and ToU of energy consumption, Figs. 10 and 11 compare the cumulative energy and electricity costs of the system between the base scenario and the optimal scenario during the day, respectively. As can be seen in both figures, the proposed optimized strategy provided significant and better performance than the base scenario. The time constant (τ) obtained, which represents the system reaching approximately 63.2% of its final value (steady state) of the GH temperature, is 0.931 hours. Fig. 12 also compares the energy consumption by component in both scenarios over 24 hours. We can clearly see that the highest amount of energy consumption during the day is related to the cooling system. However, when the ambient temperature changes, such as in the early morning or evening, the amount of heating system use is also significantly increased. As said before in the methodology section, we have used an optimal strategy to find the best weight points to minimize operational daily cost. Fig. 13 illustrates the 3D Pareto front obtained by varying the weight w_1 and $w_2 \in [0.1, 0.2, \ldots, 0.9]$ where $w_1 + w_2 = 1$. These operating regimes create a range of operational balances, from a minimum-cost regime ($w_1 = 0.1$), which is

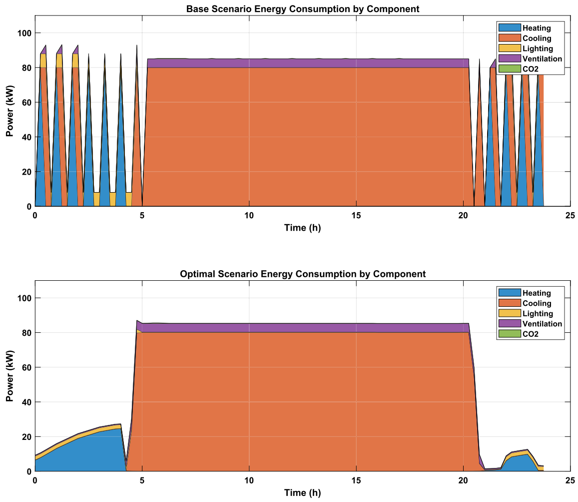

FIGURE 12. Energy consumption by component in both scenarios over 24 hours on July 1st.

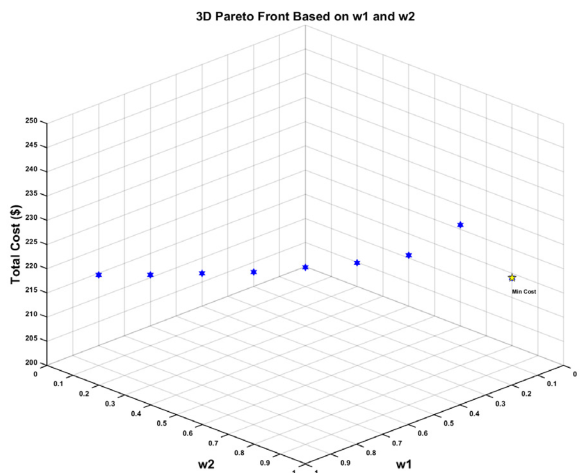

FIGURE 13. A 3-D Pareto front obtained by varying the weight w_1 and w_2 based on the final daily operational energy cost.

ideal for low demand-charge tariffs, to a balanced regime ($w_1 = w_2 = 0.5$), which can be used for cost and peak reduction vs. sensor baseline, to a minimum-peak regime ($w_2 = 0.1$), which is suitable for grid-constrained sites. Based on this, the best weights ($w_1 = 0.1$ and $w_2 = 0.9$) are chosen to minimize the final daily operational cost. In order to confirm the effectiveness and to test the performance of the optimization method, also to provide a sensitivity analysis based on changes in climate and ambient temperature, wind, and radiation, we simulate the first day of each month for every 12 months over a year based on the Regina city climate conditions. Table 3 presents the simulation results comparing the base and optimal scenarios for GH energy management over twelve representative days across the year. It shows peak demand, final cost, and percentage cost improvement for each month. The proposed strategy can achieve significant efficiency gains throughout the year, for example, validated with 43.13% cost savings and peak demand in September, compared to baseline values. The results indicate that the optimal strategy consistently reduces both peak demand and operating costs, achieving significant improvements, particularly in the summer and autumn months, while maintaining system performance. The average results confirm that the optimal scenario offers notable cost savings and efficiency gains compared to the base case. It demonstrated that with accurate next-day climatic data, we can provide an optimal strategy to reduce the day-ahead energy of a GH in a cold region.

TABLE 3. A comparison and sensitivity analysis of base and optimal scenarios across 12 different days of the months of a year.

Day of the Year	Scenario	Peak Demand (kW)	Final Cost ($)	Improvement (%)
January 1st	Base	88.00	286.29	21.26
	Optimal	85.08	225.43	
February 1st	Base	88.00	286.21	18.97
	Optimal	85.08	231.91	
March 1st	Base	88.00	288.31	8.71
	Optimal	85.20	263.20	
April 1st	Base	88.00	292.90	20.68
	Optimal	85.24	232.33	
May 1st	Base	88.00	263.71	16.87
	Optimal	85.48	219.21	
June 1st	Base	93.30	300.83	30.35
	Optimal	85.61	209.52	
July 1st	Base	93.30	292.08	25.31
	Optimal	85.30	218.13	
August 1st	Base	93.30	297.85	33.93
	Optimal	85.62	196.78	
September 1st	Base	93.30	299.63	43.13
	Optimal	85.17	170.39	
October 1st	Base	93.30	257.65	38.87
	Optimal	85.14	157.51	
November 1st	Base	88.00	252.71	34.84
	Optimal	85.11	164.67	
December 1st	Base	88.00	285.85	37.70
	Optimal	85.14	178.08	
Average	**Base**	**90.21**	**283.67**	**27.55**
	Optimal	**85.41**	**203.10**	

IV. DISCUSSION

The optimization results showcased how effective a multi-objective approach can be in striking a balance between energy consumption, peak demand, and costs in a smart GH system. The schedules created for various systems, like lighting, heating, cooling, CO_2 injection, and ventilation, reflect a thoughtful consideration of the GH's operational needs while keeping energy use and costs to a minimum. These schedules are designed to optimize energy use by turning on systems only when they're needed, such as adjusting heating and cooling based on outside temperature changes. This strategy cuts down on unnecessary energy consumption, which in turn lowers the GH's operational costs. The findings revealed a remarkable average of 27.55% energy savings a day over a year with the optimized strategy compared to the base scenario of the GH control system. The temperature profile indicates that the GH successfully maintains a stable internal environment, which is crucial for consistent plant growth. Due to well-timed operation of the heating and cooling systems, the GH temperature stays within the ideal range. Additionally, the ventilation schedule is vital for ensuring good air quality and humidity levels, further enhancing the growing environment. The optimized CO_2 schedule guarantees that plants receive CO_2 enrichment within optimal and predefined ranges, preventing waste and promoting sustainable resource use. This not only increases plant growth, but also supports the overall sustainability of GH operations. By syncing the activation of energy-intensive systems with off-peak energy pricing hours, this optimization effectively manages energy costs. It avoids peak energy costs whenever possible, scheduling operations during off-peak hours when energy prices are lower. Overall, the results of this optimization demonstrate that the GH can maintain ideal growing conditions by carefully balancing various objectives while minimizing energy consumption and operational costs. These results underscore the importance of advanced control strategies in modern agricultural systems, particularly in cold environments where resources and energy should be managed precisely.

V. CONCLUSION

This study presented a multi-objective optimization approach for controlling and managing energy consumption and peak demand based on electricity cost and time of use (ToU) of equipment in a smart GH system. The tomato crop was selected for its optimum compatibility with the environmental controls of the GH for intense growth in the provided ranges of CO_2, humidity, and temperature. The focus of the methodology on energy efficiency and control precision maximizes tomato yield and sustainability. Control schedules optimized for significant systems such as lighting, heating, cooling, CO_2, and ventilation, the research illustrates sensational improvements in operating energy efficiency and cost savings over the year. We compared the results of sensor-based and optimal scenarios to control the operating conditions of the GH to confirm the effectiveness of the proposed strategy over 12 days (first day of each month) over a year. The obtained results revealed a remarkable average of 27.55% energy savings a day over a year, and up to 43.13% or more on some days with the optimized strategy compared to the base scenario of the GH control system. This analysis provided important insight into the trade-offs that occur in reducing total energy cost and peak demand, presenting an optimal solution to reconcile these competing objectives. This reconciliation is needed to enable the GH to remain within its energy budget while ensuring that environmental conditions are adequate for maximum plant growth. The findings have highlighted the importance of advanced control systems in modern agriculture, where resource efficiency is vital both for economic sustainability and environmental supervision. Subsequent studies can explore the use of other environmental variables and real-time adaptive control to further enhance the optimization process and address dynamic issues faced by GH operators.

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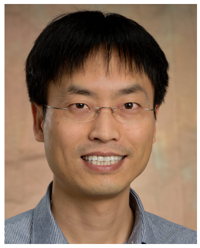

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